

Decomposition of Monolith Applications Into Microservices Architectures: A Systematic Review

Source: Y. Abgaz *et al.* "Decomposition of Monolith Applications Into Microservices Architectures: A Systematic Review," *IEEE Transactions on Software Engineering*, 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10160171/>

Paper Information: Key Paper, Survey, Systematic Literature Review (SLR)

Rating: ★★★★★ (5/5)

Own Keywords: Background, Microservices, Migration, Monolith, Monolith Decomposition

Reading Progress: First Pass

Introduction (What is the Paper about?)

- **Aim: give an overview about the field of *Monolith Decomposition***
 - includes different phases of monolith decomposition
 - identifying research gaps
 - outlining future directions
- **Comprehensive and interrelated picture of various decomposition techniques, benchmarks and metrics**
 - outlining differing migration processes
 - identify existing tooling
 - investigate contributions seeking to define migration road maps
 - problems and challenges during migration

Methods

- **Systematic Literature Review (SLR)** protocol and *snowballing* method
 - identify 5,022 search results from major computer science literature databases
 - applies four refinement steps concluding to 33 papers
 - after applying snowballing research finally ends up with 35 papers
- **Examination and Evaluation using follow-up research** after the original paper was made public

Scientific Benefit

- **Introduced the *Monolith to Microservices Decomposition Framework (M2MDF)*** which identifies the mayor phases and key elements of decomposition
 - provides a comprehensive map for researchers and practitioners
- **Detailed Analysis of existing Decomposition approaches, tools and methods**
 - monolith decomposition remains at an early stage and remains a complicated and expensive task
 - lack of integrated, comprehensive data collection and analysis methods with respect to crucial monolith aspects
 - lack of comparison between various decomposition methods
 - insufficient tool support, standardized metrics and datasets
 - absence of methods combining static, dynamic and evolutionary data
- **Identification of used metrics and datasets for Evaluation**

- **can assist practitioners seeking to understand the dimensions of monolith decomposition** and current capabilities
 - findings valuable for researches looking to identify areas of further extension

Challenges / Future Work

Limitations

Paper Outline and Read Progress

1. [X] Introduction
2. [X] Background and Related Work
3. [] SLR Methodology
4. [] Presentation of the *M2MDF*
 - Detailed Results of the SLR
5. [] Existing Gaps in the field
6. [] Threats to validity
7. [] Summary, Discussion
 - Specific implications for practitioners and researchers
 - Inclusion of post-review reflection evaluating M2MDF examining follow-up literature
8. [] Conclusion and future research directions

Own Comments

(+) Key Paper